

Persisting challenges

Fig. 1 shows a high-level IoT architecture. There is an IoT cloud platform where you store your data, telemetry, your dashboards that are pushed by the data you are storing, some analytics on the cloud, machine learning models, etc. You do all of your operational management through that cloud environment only.

On the other hand, you have edge computing with a whole lot of devices that are connected to many sensors. These devices may be PLCs, protocol gateways, and control boards with varied computation power where you can run big and small workloads. But, in general, all the IoT topologies are solving the given problems. So, what are the challenges you may wonder? Here are some, to begin with:

Cloud connections. These devices have to talk to the cloud and these connections are expensive. It is a very difficult process to manage these cloud connections on a continuous basis.

Device ops. It is generally an afterthought. Developers tend to focus more on the applications and forget about provisioning that application on a device and managing it from the device.

Why cloud connections are complex

Cloud connections are complex due to multiple reasons. Here are a few:

Security challenges. There are huge security challenges. To maintain security, you have to perform basic authentication to certificate management. You may even have security modules tied in, hence making this paradigm quite complex.

In addition, there are reliability challenges. Often you will come across a use case where you want no data lost but the connections are not permanent. The device is normally placed far in remote cor-

ners where networks are flimsy. In those cases, you need to make sure that your data is backed up on your device. Bringing in those reliability aspects for the data makes it even more challenging.

In many cases when you start to code your applications in a particular language, you tend to write your creativity logic in that same language. Hence, there is a conflict between the language that your infrastructure management is written in and the language your creative logic is written in. It becomes a huge bottleneck to software portability. Oftentimes you cannot make software that can work on all the devices.

High degree of customisation needed. We need fundamental customisation, because we know that the most fragmented space is the hardware space of the devices. There are so many different vendors, varied hardware architecture, and on top of that we have so many flavours of operating systems and operating environments that drive that hardware. Having something which runs on a device in any hardware and firmware environment is a really complicated challenge.

Limited computational resources. The width of computation of power on these devices is from a few megabytes to gigabytes. You need your infrastructure managed in a way that is cheaper and lets you save computational resources. Rolling out a continuous software update is a difficult thing to do. In addition, you just not only have a single device to handle. Generally, it is a fleet of devices where your business logic has to be rolled out, and that makes it difficult to handle.

Remote only. If you want to handle all these devices, update them, manage them, manage their firmware and software, provision your business logic on a continuous basis, all remotely, it becomes quite challenging.

Recently emerged challenges

Lately, some big companies like IBM and Google have been re-thinking their investment in IoT and they have been re-strategising their engagement in the IoT market. It means that vendor lock-ins are real. You might get tied into a vendor who goes through a churn and decides to leave the market or change its strategy.

The other part is that you do not get the freedom to choose. Multi-cloud approach has been growing really well. You as a provider of a value added service, or you as a developer who wants to bring something new, would want to use the best available cloud service. You would like to use a device capability of the best IoT platform on the cloud. But for your artificial intelligence (AI) and machine learning (ML) needs you would like to choose Amazon Web Services (AWS) to run your AI pipelines.

In many cases you would like to push your data or use the services on the device that are talking to multiple clouds. And that again is really not possible, as you have to think bottom up. If you do not use the right services, you need to really think from the ground up and pick up your technology and assess the cost of development. And the final bit, if there is a vendor that you are dealing with and you have the business continuity risk.

Hopefully, this gives you a picture of the challenges. Let us focus on the device side of the challenges and see how we can solve them on the device side. As mentioned before, every challenge brings new opportunities.

Solutions to the challenges

You have written a bunch of components, your PLCs, protocol drivers through which you are getting your data from the sensors, and you have a bunch of applications that you want to run which contain your business logic. You first need to take a synchronous way for communication